Unit: 6 Unit 2: Exponents Radicals
Topic: Operations with Radicals

Name: ________________
Date: ___________

Challenge (Habanero)

Q1. Simplify $\sqrt{50} + \sqrt{18}$
(A) $\sqrt{2} + 3\sqrt{2}$
(B) $5\sqrt{2} + 3\sqrt{2}$
(C) $7\sqrt{2}$
(D) $5 + 3\sqrt{2}$
(E) $8\sqrt{2}$

Q2. A hiker is traveling at a rate of $4\sqrt{3}$ miles per hour. How far will they travel in $\sqrt{2}$ hours?
(A) $4\sqrt{6}$ miles
(B) $5\sqrt{6}$ miles
(C) $6\sqrt{2}$ miles
(D) $8\sqrt{3}$ miles
(E) $8\sqrt{6}$ miles

Q3. Simplify $\frac{\sqrt{48}}{\sqrt{12}}$
(A) $\sqrt{4}$
(B) 2
(C) $\sqrt{2}$
(D) 4
(E) $2\sqrt{2}$

Q4. A car's fuel consumption is $\sqrt{20}$ miles per gallon. If the car travels $5\sqrt{5}$ miles, how many gallons of fuel are consumed?
(A) $\sqrt{5}$ gallons
(B) 5 gallons
(C) $5\sqrt{2}$ gallons
(D) $\sqrt{10}$ gallons
(E) $\sqrt{4}$ gallons

Q5. The population of rabbits in a forest is given by $4\sqrt{t}$ after t months. If there were 12 rabbits initially, how many will there be in $\sqrt{9}$ months?
(A) 16 rabbits
(B) 12 rabbits
(C) 20 rabbits
(D) 4 rabbits
(E) 8 rabbits

Q6. Simplify $(\sqrt{3} + \sqrt{2})^2$
(A) $3 + 2\sqrt{6} + 2$
(B) $5 + 2\sqrt{6}$
(C) $7 + 2\sqrt{6}$
(D) $1 + 2\sqrt{6}$
(E) $9 + 2\sqrt{6}$

Q7. The area of a rectangle is given by $A = \sqrt{2}l \cdot \sqrt{5}w$, where l is the length and w is the width. If the area is $10\sqrt{10}$ square units, and the length is $\sqrt{2}$ units, what is the width?
(A) 5 units
(B) $\sqrt{5}$ units
(C) $5\sqrt{2}$ units
(D) $\sqrt{10}$ units
(E) $2\sqrt{5}$ units

Q8. Simplify $\frac{\sqrt{72}}{\sqrt{8}}$
(A) $\sqrt{9}$
(B) 3
(C) $\sqrt{3}$
(D) $3\sqrt{2}$
(E) $\sqrt{2}$

Open Ended Questions (FRQ)

Q9. A park ranger travels $4\sqrt{5}$ miles in $\sqrt{5}$ hours. What is their average speed in miles per hour? Show your work and simplify your answer.

Q10. If the population of deer in a forest is modeled by $2\sqrt{t}$ after t years, and the initial population was 8 deer, how many deer will there be after 9 years? Show your work and simplify your answer.

Chef's Tips

- Remind students to simplify radical expressions
- Encourage students to check their work
- Emphasize the importance of showing calculations for free-response questions